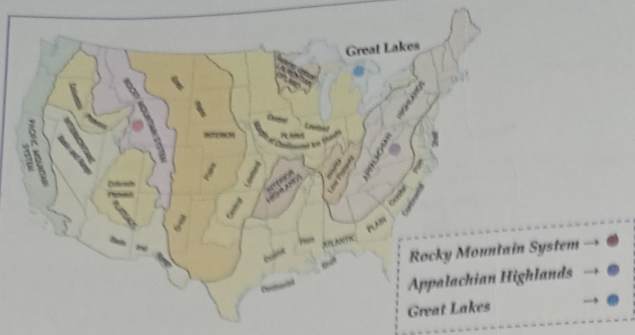




SOUTH AMERICA



Country	Capital
Chile	Santiago
Argentina	Buenos Aires
Paraguay	Asuncion
Bolivia	Sucre
Uruguay	Montevideo
Brazil	Brasilia
Venezuela	Caracas

Brazil: Currency: *Brazilian Real*.

- It is the largest country of South America.

South America has only two landlocked countries: *Bolivia* and *Paraguay*.

Mountains

Ojas del Salado: It is the *highest mountain* (dormant complex volcano) in the *Andes* range on *Argentina-Chile* border.

Mt. Cotopaxi: It is located in the Andes mountains of *Ecuador*.

Andes Mountain Range: It is the *longest chain of mountain range* on land that runs along the western coast of *South America*. It is a *young fold mountain range*.

- It passes through seven countries: *Venezuela*, *Colombia*, *Ecuador*, *Peru*, *Bolivia*, *Chile*, and *Argentina*.

Mt. Cotopaxi and *Mt. Chimborazo* are two active volcanoes located in *Ecuador*.

Deserts

Atacama Desert: It is the *driest desert* located in *Northern Chile* bordered by *Pacific Ocean* (west) and *Andes Mountains* (east).

Patagonian Desert: It is located in *Argentina*.

Lakes

Lake Titicaca: It is the world's *highest navigable lake*, situated in the Andes mountains on the border of *Peru* and *Bolivia*.

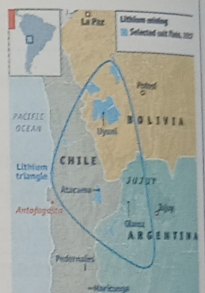
Lake Maracaibo: It is located in *Venezuela* (*South America's largest lake*). It is also one of the world's oldest (formed 36 million years ago), that's why it's rich in oil and gas resources and is known as the "*oil lake*".

Rivers and Islands

Amazon River: It is the world's *largest river by volume* and *second-longest river*, originates from *Peru*, flows through *Ecuador*, *Colombia*, *Venezuela*, *Bolivia*, and mainly *Brazil* before emptying into the *Atlantic Ocean*.

Galapagos Island: It is part of *Ecuador*, located in the *Pacific Ocean* approximately 600 miles away in the west, from *Ecuador's* coast.

The *Lithium Triangle* is a region of the *Andes* that is rich in *lithium* reserves, encompassed by the borders of *Argentina*, *Bolivia*, and *Chile*.



EUROPE



Russia: Russia is considered a part of both *Europe* and *Asia*.

Greenland: It is the *world's largest island*, located in the *North Atlantic Ocean*, and it is an autonomous territory within the Kingdom of *Denmark*.

Iceland: It is the *largest volcanic island* located in the North Atlantic Ocean.

Do you know the difference between Britain and the United Kingdom?

- Great Britain (Britain) → Island consists of England + Scotland + Wales
- United Kingdom (UK) → Political country consists of England + Scotland + Wales + Northern Ireland
- So, the United Kingdom = Great Britain + Northern Ireland.

• *Switzerland and Austria* are landlocked countries.

Balkan Nations: It refers to the countries located on the Balkan Peninsula in southeastern Europe.

Mountain Ranges

Ural Mountain Ranges: It separates *Europe* and *Asia* and is an *old fold mountain range*.

Alps Mountain Range: The Alps are the *highest mountain range* in Europe and are *young fold mountains*.

Scandinavian Countries and Nordic Countries



- Iceland →
- Norway →
- Sweden →
- Denmark →
- Finland →
- Hinderburg Line →

• Scandinavian countries: *Norway, Sweden, Denmark*

• Nordic countries: *Denmark, Finland, Sweden, Norway, Iceland*

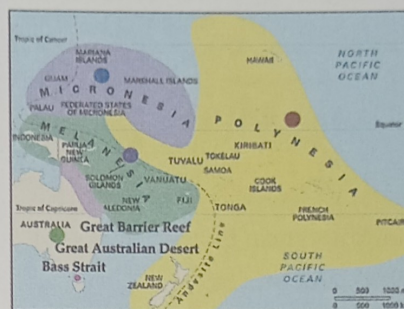
Hinderburg Line: It is a line between *Germany* and *Poland* (*German defensive line* in French territory during *World War I*).

Some famous countries of Europe	Capitals
Norway (Land of Midnight Sun)	Oslo
Sweden	Stockholm
Finland	Helsinki
Denmark	Copenhagen
Germany	Berlin
Belgium	Brussels
Poland	Warsaw
France	Paris
Spain	Madrid
Portugal	Lisbon

Italy	Rome
Greece	Athens (World's first Olympic games were held in 1896)
Romania	Bucharest
Hungary	Budapest
Bulgaria	Sofia
Czech Republic	Prague
Austria	Vienna
Switzerland	Bern

Vatican City- *Smallest country* in the world.

OCEANIA



- Melanesia →
- Polynesia →
- Micronesia →
- Great Barrier Reef →
- Great Australian Desert →
- Bass Strait →

• Oceania is made up of *Micronesia, Polynesia, and Melanesia*, in addition to *Australasia*, which includes *Australia and New Zealand*.

• *Puncak Jaya* (also known as *Carstensz Pyramid*) is the *tallest mountain* in Oceania. (4884 m) (Located in Papua New Guinea).

Melanesia: Largest island in Melanesia: *New Guinea* (*capital - Port Moresby*).

• Papua New Guinea, Solomon Islands, Santa Cruz, Fiji - are parts of Melanesia. (*Capital of Fiji - Suva*).

Micronesia: *Marshall Islands* and *Kiribati* are parts of Micronesia.

Polynesia: Samoa, Tonga, Hawaii, Easter Island, Cook Islands and New Zealand are parts of Polynesia (Hawaii Islands are parts of the USA, while Easter island is part of Chile (South America)).

Rivers

- The longest river of Australia is *Murray*.
- *Darling River* is Australia's other important river.

Straits

Bass Strait

- It separates *Tasmania* from *Victoria* in Australia.

Great Barrier Reef

- It is the world's *largest barrier reef* located in North Eastern coast of Australia.
- It is also the world's *largest coral reef system*.

Deserts

Great Australian Desert: The Great Australian Desert covers central and western

Australia.

• It is made up of several interconnected deserts, making it the *largest in Australia.*

• Some major deserts are: *Victoria Desert, Great Sandy Desert, Tanami Desert, Simpson Desert, Gibson Desert.*

Islands

- *Stewart island* and *Cook islands* are parts of New Zealand.
- *King island* is part of Australia.

- Capital of Australia- *Canberra*
- Capital of New Zealand- *Wellington (located in North Island)*
- Capital of Fiji- *Suva*

ANTARCTICA AND ARCTIC

ANTARCTICA

- Indian Research Centre in Antarctica: *Bharati, Dakshin Gangotri (established in 1984), and Maitri.*

ARCTIC

- India's *first research centre "Himadri"*, located at *Spitsbergen, Svalbard, Norway*, was established here in 2008.

Highest peaks of all Seven Continents

Mt. Everest (8848 m) - Asia

Mt. Aconcagua (6960 m) - South America

Mt. Denali/ McKinley (6190 m) - North America

Mt. Kilimanjaro (5895 m) - Africa

Mt. Elbrus (5642 m) - Europe

Mt. Vinson Massif (4892 m) - Antarctica

Mt. Kosciuszko (2228 m) - Australia

Decreasing
order in
height

CITIES LOCATED ON BANKS OF RIVER

River	Cities
Seine river	Paris
Thames river	London
Tigris river	Baghdad
Danube river	Budapest
Nile river	Cairo
Ravi river	Lahore
Moskva river	Moscow
St. Lawrence river	Montreal/Quebec

COUNTRIES LOCATED ALONG THE TROPICS

Countries through which Tropic of Cancer Passes:

1. Mexico
2. Bahamas
3. Western Sahara
4. Mauritania
5. Mali
6. Algeria
7. Nigeria
8. Libya
9. Egypt
10. Saudi Arabia
11. UAE
12. Oman
13. India
14. Bangladesh
15. Myanmar
16. China
17. Taiwan

Countries through which Equator Passes:

1. Colombia
2. Ecuador
3. Brazil
4. Sao tome and Principe
5. Republic of Congo
6. Gabon
7. DRC Congo
8. Uganda
9. Kenya
10. Somalia
11. Maldives
12. Indonesia
13. Kiribati

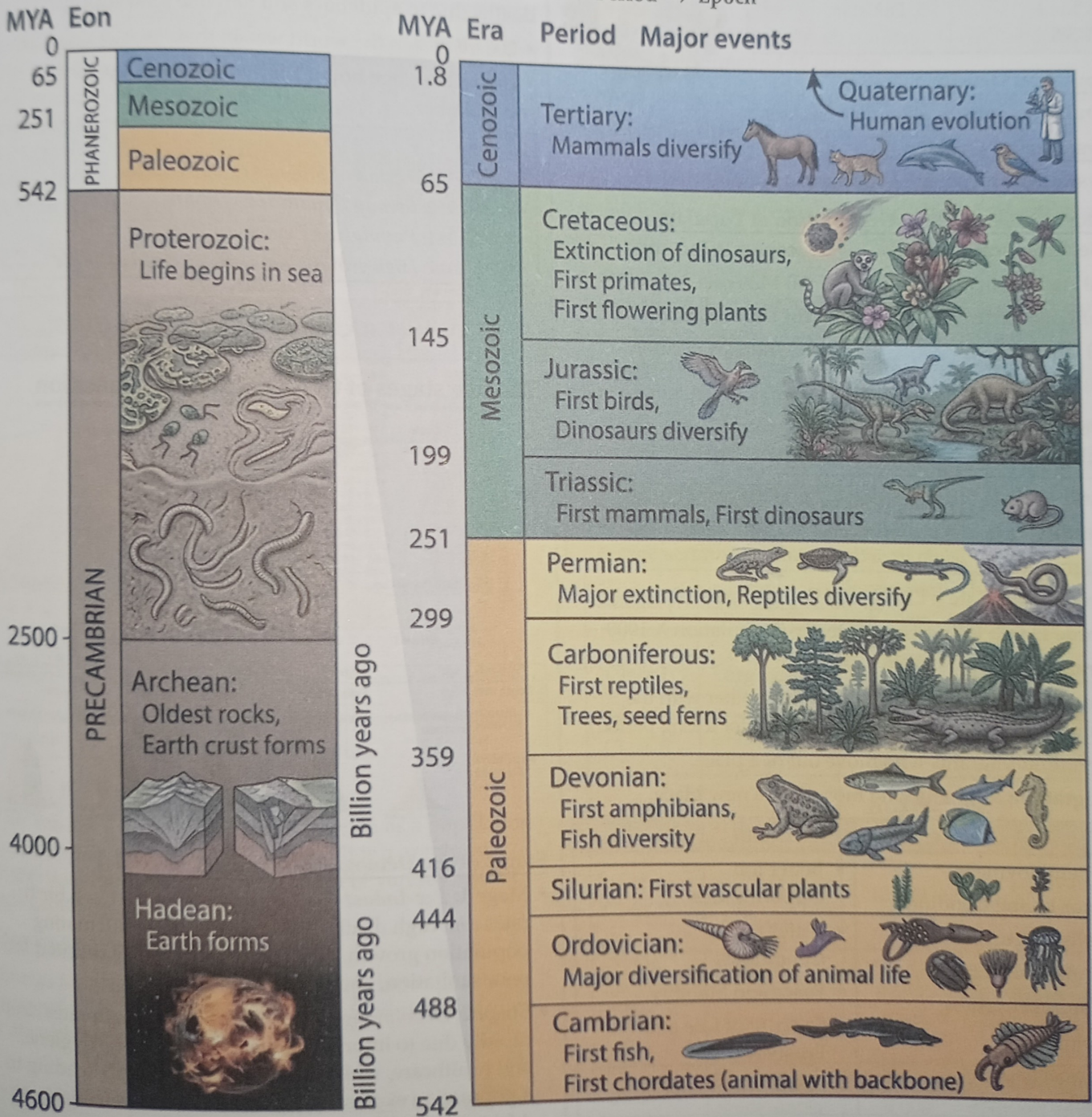
Countries through which Tropic of Capricorn Passes:

1. Chile
2. Argentina
3. Paraguay
4. Brazil
5. Namibia
6. Botswana
7. South Africa
8. Mozambique
9. Madagascar
10. Australia



Geographical Timescale

SuperEon → Eon → Era → Period → Epoch



POPULATION DISTRIBUTION

Pattern:

The top 10 most populous countries of the world contribute about 60% of the world's population in which 6 are located in Asia.

• They are:

1. India: 17.78%
2. China: 17.20%
3. Indonesia: 3.47%
4. Pakistan: 2.90%
5. Bangladesh: 2.10%
6. Russia: 1.80%

Worldwide Five most Populous Countries are:

1. India
2. China
3. USA
4. Indonesia
5. Pakistan



Population Density

Population density = Population/Area

Physiological Density

Physiological Density = Population/Arable Land

Factors Influencing the Distribution of Population

1. Geographical Factors
 - (a) Availability Of Water
 - (b) Landforms
 - (c) Climate
 - (d) Soil
2. Economic Factors
 - (a) Minerals
 - (b) Urbanisation
 - (c) Industrialisation

3. Social and Cultural Factors

Components of Population Change

1. Birth
2. Death
3. Migration

India's crude birth rate (CBR) has declined from 19.1 in 2022 to 18.4 in 2023, according to data from the Sample Registration System.

- Crude Birth Rate = Live Birth/ Population X 1000
- Crude Death Rate = No. of Deaths/ Population X 1000

Migration:

Movement of people from one place to another with the intention of settling there (permanently or for a long period).

Emigration: Migrants who move out of a place.

Immigration: Migrants who move into a new place.

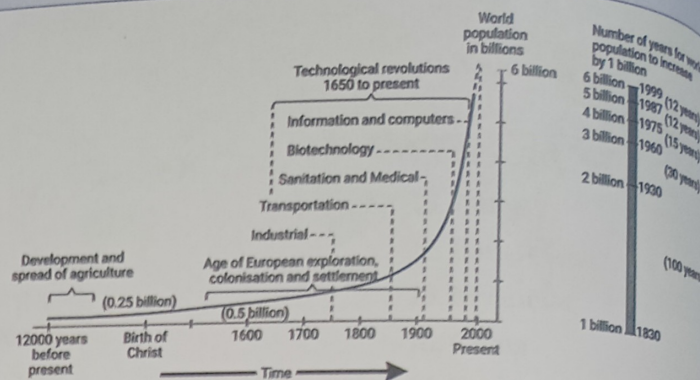
Push Factors

- Unemployment,
- Poor Living Condition,
- Political Turmoil,
- Unpleasant Climate,
- Natural Disasters,
- Epidemics,
- Socioeconomic Backwardness.

Pull Factors

- Better Job Opportunities,
- Better Living Conditions,
- Peace and Stability,
- Security of Life and Prosperity,
- Pleasant Climate.

TRENDS IN POPULATION GROWTH IN INDIA



- The first stage has a high Birth Rate and Death Rate because people reproduce more to Compensate for the deaths due to epidemics and variable food supply.
- 200 years ago the world was in this stage.
- In the last stage both fertility and mortality decline considerably.

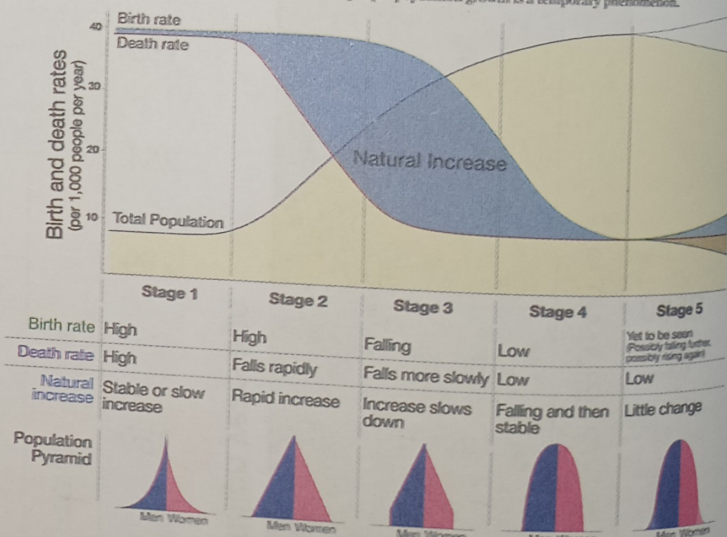
In India

- 1901-1921: Period of Stagnant Population
- 1921-1952: Steady Population Growth
- 1951-1981: Population Explosion
- 1981-2011: High growth but signs of slowing down

DEMOGRAPHIC TRANSITION MODEL

The five stages of the demographic transition

The demographic transition is a model that describes why rapid population growth is a temporary phenomenon.



Stages of the Demographic Transition Model

- **Stage 1: Pre-Industrial (High Stationary):** High birth rates and high death rates result in slow, fluctuating population growth. High mortality is due to limited food, poor sanitation, and disease.
- **Stage 2: Developing (Early Expanding):** Death rates drop rapidly due to improvements in food supply, hygiene, and healthcare, while birth rates remain high, leading to a population explosion. Eg. include many sub-Saharan African nations.

- **Stage 3: Industrializing (Late Expanding):** Birth rates begin to decline due to urbanization, increased education (especially for women), and better access to contraception. Population growth starts to slow. Eg. India.
- **Stage 4: Developed (Low Stationary):** Both birth and death rates are low, stabilizing the population. Improved living standards and smaller family sizes are common. Eg. the USA and UK.
- **Stage 5: Post-Industrial (Declining):** Birth rates fall below death rates, leading to a natural decrease and an aging population. Eg. Japan, Germany, and Italy.

Population Control Measures:

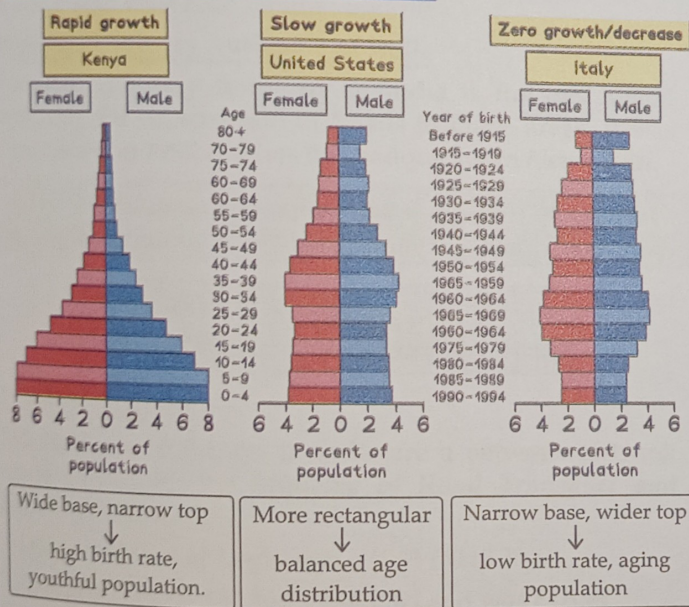
- Access to Family Planning Services
- Free availability of contraceptives

National Family Planning Programme:
The National Family Planning Programme was launched in 1952 in India during the 4th Five Year Plan, making it the world's first national program of its kind. Its initial focus was on reducing birth rates to stabilize population growth.



जोड़ी जिम्मेदार
जो प्लान करे परिवार

Three Shapes of Population Pyramid



Age Sex Pyramid:

- An age-sex pyramid is primarily known as a population pyramid or an age structure diagram. It is a graphical illustration that shows the distribution of various age for each gender in a specific population.

Thomas Malthus Theory:

Thomas Malthus's theory states that population grows at a geometric rate (multiplying) while the food supply grows at an arithmetic rate (adding).

OR,

The number of people would increase faster than the food supply.

Any further increase would result in a population crash caused by famine, disease and war.

Population Composition:

- **Working Population (Demographic Dividend):** Age 15-59 Yrs.

SETTLEMENTS

Types of Settlements:

1. Urban Settlement:

- **Towns:** Urban settlements with a population of *less than 100,000*.
- **Cities:** Urban settlements with a population between *100,000 and 1 million*.
- **Metropolitan Cities:** Urban settlements with a population between *1 million and 10 million*.
- **Mega Cities:** Cities with a population exceeding *10 million*.
- **Census Towns:** Settlements that exhibit urban characteristics (like having a population of 5,000 or more) but are not governed by a statutory local body like a municipality; they are governed by rural administrative structures.

2. Rural settlements:

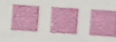
- **Compact/Clustered/Nucleated:** Houses are built closely together in a cluster.
- **Semi-compact/Semi-clustered/Fragmented:** A cluster of houses that may be divided into several sub-clusters.
- **Hamleted:** Consists of one or more hamlets scattered over a wider area, often at a distance from the main village.
- **Dispersed:** Houses are widely separated from one another.

Types of Class Town:

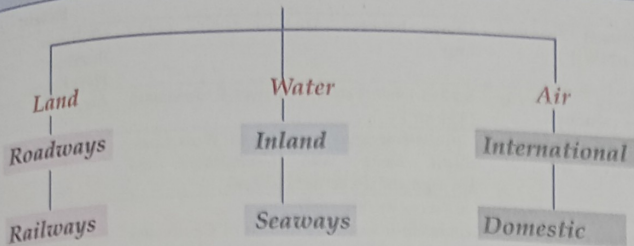
Category	Type of Town/City	Population Range	Example
Class I	Large City	1,00,000 and above	Jaipur, Lucknow, Coimbatore
Class II	Medium Town	50,000 - 99,999	Saharanpur, Bhillai, Siliguri
Class III	Small Town	20,000 - 49,999	Satna, Ambikapur, Bhiwadi
Class IV	Very Small Town	10,000 - 19,999	Lonavala, Hoshiarpur (earlier stages)
Class V	Semi-Urban Area	5,000 - 9,999	Tirur, Balurghat (smaller urban centres)
Class VI	Transitional Settlements	Less than 5,000	Emerging urban villages

The Census of India (under the Ministry of Home Affairs) defines urban areas for demographic and statistical purposes, while the Ministry of Housing and Urban Affairs (MoHUA) is responsible for urban planning, governance, and policy implementation.

This system helps in Urban planning, allocation of resources, and implementation of Government schemes.



MODES OF TRANSPORT



MODES OF TRANSPORT

Roadways

- India's road network is the **second largest** in the world.
- Total road network length : **66.71 lakh kilometres**

Detailed Breakdown:

- Total National Highway Length: **1,46,145 km**.
- State Highways: **1,79,535 km**.
- Expressways: **5,579 km** (as of July 2023).
- Highest road density : **Kerala**
- Lowest road density : **Arunachal Pradesh**
- State with maximum road length : **Maharashtra**
- State with least road length : **Sikkim**

National Highway Authority of India

National Highway Authority of India is the designated authority for construction of Roads in Plane areas. It was established in 1995 and has its headquarter in **New Delhi**.

Public Works Department(PwD) established in 1854 works for the maintenance of roads at the district and village level.

Motor Vehicle Act: 1988.

National Highway Authority Act: 1988.

National Highways

The National Highways in India are a network of trunk roads owned by the **Ministry of Road Transport and Highways (MoRTH)**.

- Longest National Highway: **NH 44 (4112 Km)**. It connects **Srinagar to Kanyakumari** and was earlier known as **NH 7**.
- Shortest National Highway: **NH 548 & NH 118 (5Km)**

The historically important Sher Shah Suri Marg is known as National Highway 1. It links Delhi and Amritsar.

Bharatmala Pariyojana:

The Bharatmala Pariyojana was launched with the primary focus on optimizing the efficiency of the movement of goods and people across the country.

- Phase I-approved in October 2017, focuses on **development of National Highways**.

Note: The Sagarmala Project, launched in 2015, is related to the development of ports.

- Bhupen Hazarika Bridge** is the **longest** bridge constructed over Lohit river and connects Assam to Arunachal Pradesh.

The Golden Quadrilateral (GQ)

- The **Golden Quadrilateral(GQ)** project initiated in **1999**, is a national highway network connecting most of the major industrial, agricultural and cultural centres of India.
- It forms a quadrilateral connecting the **four** major metro cities of India, viz., **Delhi (north), Kolkata (east), Mumbai (west) and Chennai (south)**.

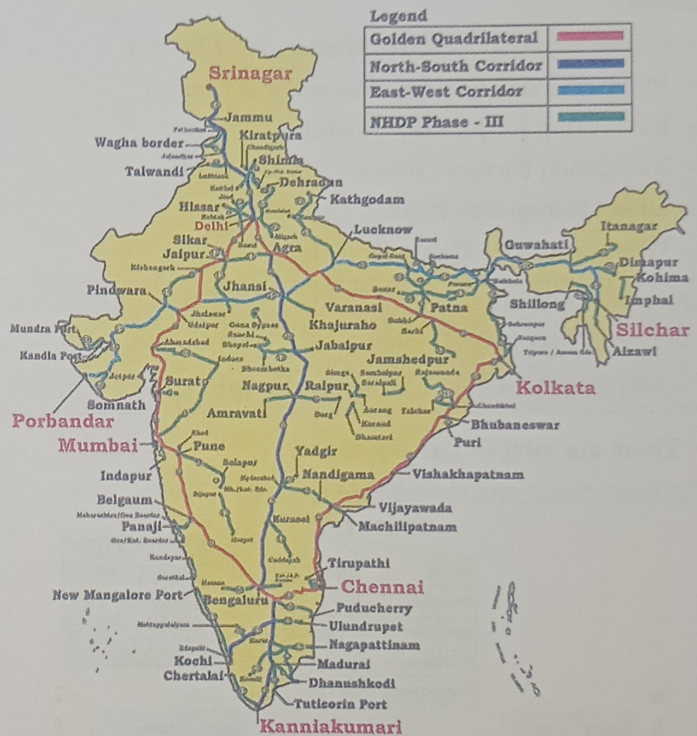
Delhi-Kolkata: NH19 (erstwhile NH2)

Kolkata-Mumbai: NH46 (erstwhile NH6)

Mumbai-Chennai: NH48 (erstwhile NH4)

Delhi-Mumbai: NH48 (erstwhile NH8)

- It is a system of highways that spans a total length of **5,846 Km (3,633 miles)**.



North-South-East-West Corridor

The North-South and East-West Corridor is a network of national highways connecting major cities of India. It consists of **two highway systems** - the North-South highway extending 4,000 km from Srinagar in the north to Kanyakumari in the south with a spur to Kochi and the East-West highway extending 3,300 from Silchar in the east to Porbandar in the west.

- Jhansi** is the **junction** of North-South and East-West Corridors.

Border Roads

The Border Roads Organisation (Ministry of Defence), established in **1960**, aims to boost economic development and defense preparedness by improving road infrastructure in northern and **northeastern** border areas.

Railways

Significant facts about Indian railway network:

- Indian Railways is the **4th largest** railway network in the world.
- Dibrugarh (Assam) to Kanyakumari (Tamil Nadu)** is the **longest train route** in India & Vivek Express runs on this route.
- Uttar Pradesh** has the **longest route of Railway Lines in India** while Rajasthan is at **2nd Place**.
- Longest Rail Bridge: Bogibeel Bridge (4.94 Km)**; constructed on the Brahmaputra river - connects Dhemaji & Dibrugarh
- Highest Rail Bridge: Chenab Bridge (359m)**
- Longest Railway Station: Hubballi (Karnataka) (1.5 Km)**
- 2nd Longest: Gorakhpur**

In India Railway Coaches are Manufactured in following places:

- Chittaranjan Locomotive Works : **West Bengal**
- Integral Coach Factory : **Perambur (Chennai)**
- Rail Spring Karkhana : **Gwalior**
- Rail Coach Factory : **Kapurthala (Punjab)**
- Diesel locomotive Works : **Varanasi**
- Wheel and Axle Plant : **Bengaluru**

1st train in India :

- Introduced by Dalhousie.
- ran from Mumbai to Thane on 16 April 1853.

- There are total **18** railway zones.

Railway zones and their headquarters

NORTHERN		RAILWAY ZONES & HEAD-QUARTERS	
Northern	New Delhi		
North Western	Jalpur	Eastern	Kolkata
North Central	Prayagraj	East Central	Hajipur
North Eastern	Gorakhpur	East Coast	Bhubaneswar
North East Frontier	Guwahati		
Western	Mumbai		
West Central	Jabalpur		
Central	Mumbai		
Southern	Chennai		
South Western	Hubballi		
South Central	Secunderabad		
South Coastal	Visakhapatnam		
South East Central	Bilaspur		
South Eastern	Kolkata		
SOUTHERN			

WATER TRANSPORT

Nearly **90%** of India's trade Volume (in terms of value), is moved by sea making shipping, the backbone of trade and economic growth.

1st Under Water Metro : Kolkata
Metro Man of India : E. Sreedharan

National Waterways

Series Number	Name and Stretch	Geographic Extent
*National Waterway No.1(NW1)	Ganga between Allahabad-Haldia (1620 km)	UP, Bihar, Jharkhand and West Bengal
National Waterway No.2 (NW2)	Sadiya-Dhubri stretch of river Brahmaputra (891 km)	Assam
National Waterway No.3(NW: III)	Kollam-Kottapuram stretch of West Coast Canal along with Champakara and Udyogmandal Canals (205 km)	Kerala.
Two proposed Waterways (recently declared)		
National Waterway No.4(NWIV)	Kakinada-Puducherry stretch of Canals and the Kaluvelly Tank, Bhadrachalam - Rajahmundry stretch of River Godavari and Wazirabad - Vijayawada stretch of River Krishna (1095 km)	Andhra Pradesh
National Waterway No. 5(NWV)	The Talcher-Dhamra stretch of river Brahmani, Geonkhali-Charbatia stretch of East Coast Canal, Charbatia-Dhamra stretch of Matai river and Mangalgadi-Paradip stretch of Mahanadi delta rivers (623 km).	Dhamra, Orissa

National Waterways

TOTAL LENGTH:
4,503 KM

TOTAL STATES SERVED
15



****To read more in detail about National Waterways, please refer to the 'Ports' sections**

Inland Waterways Authority of India (IWAI),

IWAI, established in **1986** under the **Ministry of Ports, Shipping and Waterways**, is responsible for the development, maintenance, and regulation of inland waterways for shipping and navigation in India. It has its headquarters in **Noida, Uttar Pradesh**.

AIR TRANSPORT

- Air services in India started in: **1911**
- State with Most no. of International Airports: **Uttar Pradesh** (Kerala stands at 2nd Place)
- Largest Airport: Indira Gandhi International Airport, IGIA (New Delhi)**

Recently in February 2025, IGAI became the first Indian airport to run entirely on solar and hydro power.

- Smallest Airport: Baljek AirPort (Meghalaya)**

- Oldest AirPort: *Juhu Aerodrome in Bombay* (established in 1928)

Airports Authority of India (AAI)

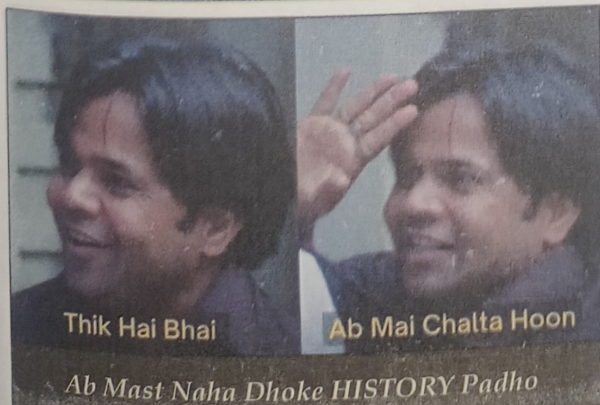
AAI is a statutory body under the ownership of the Ministry of Civil Aviation, Government of India.

- AAI currently manages a total of *137 airports*, including 34 international airports, 10 Customs Airports, 81 domestic airports, and 23 Civil enclaves at Defense airfields.
- It was established in *1995* and has its Headquarters in *New Delhi*.



Greenfield Airports: Airports that are built from scratch in a previously undeveloped land.

Brownfield Airports: These are existing airports that are upgraded, modernized, or expanded rather than built from scratch.



Thik Hai Bhai

Ab Mai Chalta Hoon

Ab Mast Naha Dhoke HISTORY Padho